

ANKITA JAIN

Staff Tech Lead | Data Engineering and AI Systems

+1 704-904-6822 • ankita.utd@gmail.com • [linkedin.com/in/jainankita](https://www.linkedin.com/in/jainankita)

SUMMARY

Staff-level Data/AI/SWE Engineering leader with 12+ years architecting large-scale distributed data and ETL platforms, and enterprise-grade Generative AI evaluation systems, at Microsoft and Bank of America. Proven track record building the data pipelines, governance frameworks, and infrastructure that feed and power production ML models and agentic AI systems at petabyte scale — from credit risk scoring to enterprise AI quality evaluation. Blends deep hands-on technical architecture (Spark, distributed data platforms, cloud-native microservices, LLM evaluation) with people leadership — hiring, mentoring, and setting technical roadmaps for cross-functional engineering teams. Experienced managing the full data lifecycle that ML and AI systems depend on: ingestion, transformation, quality, lineage, and governance.

CORE QUALIFICATIONS

- 12+ years software/data engineering; 6+ years in technical leadership and team management roles
- Deep expertise in distributed data processing: Spark, PySpark, Hadoop, Hive, Impala, Airflow, event streaming
- Built and led production agentic AI and LLM evaluation platforms (Copilot Studio, Agent Builder), with Responsible AI guardrails and observability at scale
- Cloud platforms: Azure (6+ years), large-scale cloud-native microservices on Kubernetes
- Built ETL and data pipelines that fed production ML models and enterprise AI evaluation systems — credit risk scoring, personalization, and AI quality evaluation
- Led data quality, lineage, and governance frameworks for 300M+ enterprise data assets under regulatory compliance (GDPR/CCPA)
- Technical leader-manager hybrid: architecture ownership plus hiring, mentoring, and career development for engineering teams
- 7+ years conducting technical interviews; authored technical guides and trained 50+ associates at Bank of America, plus enabled adoption across 30+ teams at Microsoft

EXPERIENCE

Lead AI and Data Platform Engineer — D365 Sales AI Microsoft

Mar 2020 – Present
Charlotte, NC

- **Data Platform & Team Leadership:** Lead a team of 6 engineers spanning data platform, evaluation, and agent engineering; serve as lead technical advisor to senior leadership on Enterprise AI Quality Evaluation Strategy, setting the technical roadmap and data standards adopted across the organization.
- **AI Evaluation Data Pipeline Architecture:** Designed and led end-to-end offline and online evaluation data platforms — ingestion, rubric-based scoring, real-time dashboards — integrated into CI/CD as production release gates and adopted by 30+ cross-functional teams to validate ML model and agent quality before launch.
- **Production Data Serving at Scale:** Shipped an online evaluation system running as a background service that continuously scores live model/agent outputs against ground-truth and heuristic signals, surfacing quality regressions in real time without impacting user-facing latency.
- **Cross-Functional ML Collaboration:** Led collaboration with the Microsoft Copilot Studio Evaluation Data Science team — driving requirements from 1P Dynamics use cases, then becoming an early evaluator of their Online Evaluation capability and shaping it ahead of broader rollout through sustained cross-timezone coordination with an Israel-based team. The resulting solution has since been adopted as-is by multiple other Microsoft 1P teams for their own Agent Quality evaluation needs.
- **Hiring & Technical Mentoring:** Conducted technical interviews for 7+ years, actively participating in hiring and candidate evaluation for engineering roles; authored technical guides and trained engineers across 30+ cross-functional teams on evaluation standards, rubric-based scoring, and CI/CD integration for the AI Quality Evaluation Framework.
- **Agentic AI & Responsible AI Governance:** Led design and delivery of production-grade agentic AI systems (Copilot Studio, Agent Builder), shipping multiple autonomous agents to General Availability and de-risking product direction through 20+ rapid POCs trading off quality, latency, safety, and cost; defined Responsible AI guardrails — real-time policy enforcement, jailbreak defenses, cross-domain grounding — sustaining production SLOs of ≥95% groundedness and ≥90% usefulness.
- **Data Governance & Quality at Scale:** Architected orchestration microservices and bulk asset-enrichment pipelines for Azure Purview on Kubernetes, lifting data throughput 50% across 300M+ enterprise assets; extended the platform with automated PII/financial data classification, sensitivity labeling, and lineage tracking to enforce access controls and regulatory compliance across the enterprise data estate.

- **Purview Cold-Tier Analytics Data Platform:** Conceptualized and led engineering (Team of 5) of an Offline Tier Analytics capability for Azure Purview, processing 1B+ daily events (~1TB/day) via multi-stage, cross-tenant orchestration — cutting compute and storage costs 50% while maintaining GDPR/CCPA-grade classification and auditing for downstream governance dashboards.
- **Observability & Reliability:** Built the alerting and observability system for the Purview Cold Tier pipeline ahead of GA, giving engineering and business teams real-time visibility into pipeline health and enabling proactive issue detection prior to customer impact.
- **Customer Data Unification:** Engineered customer-profile unification pipelines for Dynamics 365 Customer Insights (Azure, C#, Cosmos DB, PySpark) powering mapping, deduplication, matching, and merge stages to build a 360-degree customer view from disparate source systems.
- **Agent Tooling & Productivity:** Built MCP-powered tools and Claude skills for internal engineering teams, boosting engineering productivity by 45%.

Senior Data Engineer — Enterprise Credit Risk

Feb 2016 – Mar 2020

Bank of America

Charlotte, NC

- **ETL Pipeline Development for Credit Risk ML Models:** Built and operated automated data pipelines feeding production ML models for retail credit card default prediction and wholesale corporate early-warning/bankruptcy risk scoring — engineering the feature-ready data layer consumed directly by the model teams.
- **Framework Adoption at Scale:** Led adoption of Bank of America's open-source Spark ETL framework — including built-in data lineage and quality validation — across Credit Risk Platform Org, cutting manual reporting workloads by 65% enterprise-wide while maintaining financial governance compliance.
- **Legacy Modernization:** Migrated legacy Netezza pipelines to Hive, Impala, and Spark in a regulated environment; optimized Scala Spark queries to cut core ETL processing time by 50% while preserving audit-trail and compliance requirements.
- **Technical Enablement:** Authored architecture playbooks and trained 50+ associates across 5 teams to scale Hadoop/Spark adoption org-wide.
- **Regulatory Risk Data Aggregation:** Engineered Spark-SQL pipelines to securely aggregate wholesale credit risk and corporate loan data, ensuring integrity and traceability for downstream regulatory reporting.

Data Scientist

Sep 2015 – Feb 2016

Tresata, Inc.

Charlotte, NC

- Analyzed unstructured mobile app usage logs using Spark to extract behavioral and engagement metrics for client product teams.
- Cleaned, transformed, and evaluated a leading grocery retailer's consumer transaction data (Scala/Scalding), translating raw data into business recommendations.
- Built an early-stage product recommendation prototype applying statistical and ML techniques to consumer purchase data.

TECHNICAL SKILLS

- **Distributed Data & ML Infrastructure:** Spark, PySpark, Hadoop, Hive, Impala, Airflow, Event Streaming, Cosmos DB, Data Lineage, Data Quality & Governance
- **Cloud & Architecture:** Azure, Cloudera, Cloud-Native Microservices, Kubernetes, Infrastructure as Code (ARM Templates), CI/CD Pipelines
- **Programming Languages:** Python, PyTorch, Java, Scala, C#, SQL
- **AI/ML Platform:** ML Evaluation & Observability, RAG Systems, Agentic Workflows, Model Metrics Benchmarking, Responsible AI Governance
- **Leadership:** Team Building & Mentoring, Technical Roadmap & Strategy, Cross-Functional & Cross-Timezone Collaboration, Hiring

PATENTS

Dynamic Data Transformation System

US Patent 16/787,579 — Issued Aug 12, 2021

System for dynamically transforming data with built-in governance: automated data quality checks and controls at ingestion, rule-based transformations (calculations, derivations), and governance capture at each stage before publishing to downstream databases for enterprise applications.

EDUCATION

University of North Carolina, Charlotte, NC

Aug 2015

Master of Science in Data Science | GPA: 4.0